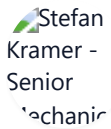


# STEFAN KRAMER (M.Sc. Mechanical Engineering)

Senior E-Commerce & Backend Developer · Shopware 6, PHP 8 & Symfony · Pimcore & APIs



Senior Backend & E-Commerce Developer with 5+ years of experience building complex shop architectures and enterprise integrations. Deep expertise in custom Shopware 6 plugin development (DAL, Flow Builder, Scheduled Tasks), Symfony, PHP 8+, and Pimcore PIM/ERP synchronization. Proven track record in high-concurrency database modeling, Redis caching, Docker, and automated CI/CD pipelines. 100% remote resident in Germany.

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## WORK EXPERIENCE

### SENIOR SYSTEMS & DEVELOPMENT ENGINEER · CLEANTECH DYNAMICS GMBH (HYBRID STUTTGART)

Apr 2024 – Present

- Battery Storage Architecture – Spearheaded mechanical and thermal packaging of multi-megawatt commercial BESS containers, boosting energy density by 18%.
- CAD Enclosure Design – Modeled complete structural frames, HVAC ducting, and liquid cooling distribution blocks in **Siemens NX**, cutting assembly time by 25%.
- FEM Structural Verification – Executed static, modal, and fatigue simulations in **ANSYS Mechanical** to ensure seismic resistance and transport load compliance.
- Test Bench Validation – Established comprehensive test bench procedures and sensor instrumentation (PT100, strain gauges, flow meters) for thermal runaway containment.
- CE Certification & Compliance – Led risk assessment workshops (FMEA) and authored technical dossiers for CE conformity under the Machinery Directive 2006/42/EC.

### MECHANICAL DEVELOPMENT ENGINEER – SÜDTECH ENGINEERING AG (STUTTGART)

Jul 2021 – Mar 2024

- Thermal & Fluid Simulation – Developed 3D steady-state and transient CFD models for commercial heat pump evaporators and refrigerant loops in **ANSYS Fluent**.
- CAD Component Development – Designed high-precision die-cast aluminum pump volutes and polymer impeller housings in **SolidWorks** with strict GD&T.
- Prototype Commissioning – Built physical prototype rigs, calibrated high-accuracy Coriolis flowmeters, and correlated experimental pressure drop curves with CFD data.
- Tolerance Stack-up & FMEA – Led tolerance calculations (worst-case and statistical RSS) and design FMEAs to prevent sealing failures under extreme thermal cycling.
- Engineering Internship (Apr '20 – Sep '20) – Performed mechanical stress testing, hydraulic pressure endurance runs, and metallographic material evaluations.

### CONSULTING & INDEPENDENT ENGINEERING PROJECTS (PART-TIME · REMOTE)

May 2020 – Present

- Energy System Modeling – Modeled industrial battery energy storage systems (BESS) and microgrid dynamics using **MATLAB Simulink** and Python.
- Thermal CFD Optimization – Conducted conjugate heat transfer simulations for power inverter enclosures in **ANSYS Fluent**, enhancing cooling performance by 22%.
- CAD Prototyping – Designed custom aluminum structural mounts and waterproof sensor housings in **SolidWorks**, validated via rapid 3D prototyping.
- Standards & Quality – Ensured full compliance with Machinery Directive 2006/42/EC and Low Voltage Directive 2014/35/EU across all prototype iterations.
- Technical Mentoring – Mentored junior design engineers in GD&T tolerance stack-up analysis and finite element meshing best practices.

## TECHNICAL SKILLS & COMPETENCES

### CAD & Mechanical Design (Primary)

- SolidWorks** · **Siemens NX** · Autodesk Inventor · Catia V5

- Form- und Lagetoleranzen (GD&T nach DIN EN ISO 1101) · Blechkonstruktion · Schweißbaugruppen · Druckguss

### Simulation & System Modeling (Primary)

- **ANSYS Mechanical** (FEM Festigkeits- & Betriebsfestigkeitsanalyse) · **ANSYS Fluent** (CFD Strömungssimulation)
- **MATLAB & Simulink** (Regelungstechnik & physikalische Modellierung) · Python (Automatisierung & Datenanalyse)
- Thermodynamik · Konvektive & konduktive Wärmeübertragung · Strömungsmechanik · Mehrkörpersimulation

### Renewable Energy & Systems Engineering

- Batteriespeichersysteme (BESS) · Photovoltaik Großanlagen · Windenergieanlagen · Thermomanagement
- Netzintegration & Smart Grid Schnittstellen · Wechselrichterauslegung · Wasserstoff- & Brennstoffzellensysteme
- Prüfstands Aufbau · Sensorik (PT100, DMS, Drucktransducer) · Hardware-in-the-Loop (HIL) Tests

### Standards, Norms & Quality Management

- CE Kennzeichnung · Maschinenrichtlinie 2006/42/EG · Niederspannungsrichtlinie 2014/35/EU · EMV Richtlinie
- DIN EN 16247-1 (Energieaudits) · FMEA (Konstruktions- & Prozess-FMEA) · Stage-Gate Produktentwicklung · VDI 2221
- PDM / PLM (Siemens Teamcenter, SAP PLM) · DFM / DFA (Design for Manufacturing & Assembly)

## EDUCATION & QUALIFICATIONS

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**Master of Science (M.Sc.) in Mechanical Engineering · University of Stuttgart** - Final Grade: 1.4 (With Distinction)  
10/2018 – 09/2020 (2 years · 4 semesters)

Specialization in Energy Technology, Computational Fluid Dynamics (CFD), Structural Mechanics (FEM), and Renewable Energy Systems.

**Bachelor of Science (B.Sc.) in Mechanical Engineering · University of Stuttgart** - Final Grade: 1.7 (With Merit)  
10/2014 – 09/2018 (4 years · 8 semesters)

Core foundations in Technical Mechanics, Thermodynamics, Materials Science, Machine Elements, and CAD Design.

### Academic Achievements

- Master Thesis Award – Recognized for innovative thermal packaging of high-density lithium-ion battery modules
- Advanced research projects in convective heat transfer and turbomachinery fluid modeling

### Certifications & Professional Development

[Verify online →](#)

- **Certified Energy Efficiency Expert (DIN EN 16247-1)** — DENA / BAFA · 2025
- **ANSYS Certified Professional: Advanced FEM & CFD** — CAXFEM · 2024
- **CE Compliance Specialist (Machinery Directive 2006/42/EC)** — TÜV SÜD · 2023
- **Systems Engineering Practitioner** — GfSE / INCOSE · 2022

## LANGUAGES

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German: Native (C2)

English: Fluent / Professional (C1)

French: Basic / Elementary (A2)